



Team Presentation Training

Abdul Rafay · Moiz · Areeba · Aqib — who says what, handovers, Q&A routing and drills.

Team
4 presenters

Deck
16 slides · ~7 min

Event
**Alibaba AI
Hackathon 2026**

How to use this book

This book trains the whole team to present the **16-slide team deck**. Every member reads the whole book, not only their own part.

Member	Slides	Extra job
Abdul Rafay	1–5, 7, 11, 13, 15, 16	Moderator (منتظم) — gives each question to the right person
Aqib	6 (demo), 12	Demo driver — clicks and fixes problems
Moiz	8	Technical backup
Areeba	9, 10, 14	Timekeeper (وقت دیکھنے والی) — signals at 1 min and 30 s left

HANDOVER LINES (باری دینا)

Rafay → Aqib: *“Aqib, please show them.”* · Aqib → Moiz: *“How does the AI decide this? Moiz.”* · Moiz → Areeba: *“A fast model is useless if it gives wrong alarms. Areeba.”* · Areeba → Rafay: *“Rafay will show how this works for a whole city.”*

Slide-by-slide script

01

0:30

Title + team All

"We are team VisionGuard. We make normal CCTV cameras smart. Let us introduce ourselves."

Each member says one line (5 s), stepping forward a little.

02

0:20

Team All

"I am Abdul Rafay, team lead... I am Moiz, ML engineer... I am Areeba, data scientist... I am Aqib, I made the website."

Each person speaks for their own card. Same order every time.

03

0:40

Problem Abdul Rafay

"Karachi has more than 20 million people and thousands of cameras. But cameras only record — nobody is really watching. An operator gets tired after about 20 minutes. So the camera sees the crime, but nobody acts."

Stop for 2 seconds after "nobody acts". Look at the judges, not the screen.

04

0:30

Solution Abdul Rafay

"VisionGuard makes cameras smart. It watches all cameras together, finds the danger, gives a risk score, saves proof, and sends the alert to the right department. Detect, decide, dispatch."

Point to the three boxes one by one while saying the three words.

05

0:20

12 detectors Abdul Rafay

"We cover twelve kinds of emergency — from fire and fights to accidents, crowds and bags left alone."

Do not read all twelve. Say 3–4, then move on.

06

2:00

Live demo Aqib

"Let me show you live. This is the control room... Here a fire is found — it goes to Fire Brigade 16... Here a fight — see the body points... Every event gets a risk score... and evidence is sealed with a fingerprint."

Switch to the app. Tell judges where to look. If slow: "While it loads, let me explain..."

07

1:00

How it works Abdul Rafay

"One frame goes in: YOLO finds objects, ByteTrack gives each person an ID, special models check pose, fire and weapons, and the event engine checks rules over time — not one frame."

Use your hand to follow the arrows left to right.

08

0:40

Models Moiz

"Moiz used transfer learning — we took YOLOv8, which already knows general shapes, and trained it more on fire, weapon and violence data on Colab GPUs."

Give Moiz credit by name.

09

0:30

Data quality Areeba

"Areeba scanned sixteen thousand fire images and seven thousand weapon images, removed duplicates, made four thousand weather images, and tuned three settings from real false alarms."

Say "data-driven" — we changed settings because of evidence.

10

0:40

Results Areeba

"Our fire model is two times better than the old colour method — precision 47.6 percent. The dataset is hard, with tiny fires, and our time checks catch fires across many frames. Weapon and violence reports are in progress."

Be honest and calm. Never round numbers up.

11

0:30

Performance

Abdul Rafay

"Our async design made it 43 percent faster on a small 4 GB GPU, and dropped half the lagging frames."

Say "measured on our test GPU".

12

0:30

Operator experience

Aqib

"Aqib built the command center: live alerts sorted by risk, voice control, heatmaps, and an evidence locker."

Give Aqib credit by name.

13

0:20

Routing

Abdul Rafay

"Each event goes to the right number: fire to 16, police events to 15, accidents to Edhi 115, crowds to 1122."

Keep it short.

14

0:30

Trust

Areeba

"We are behavioural, not biometric — no face recognition. AI suggests, a human decides, and evidence cannot be secretly changed."

Slow down here; ethics matters to judges.

15

0:30

Roadmap

Abdul Rafay

"We know our limits: fire numbers can improve, two reports are pending. Next: more training, local Karachi data, and a city pilot."

Saying limits first builds trust (اعتماد).

16

0:15

Close

All

"Cameras are already there. VisionGuard gives them a brain. Built by Moiz, Areeba, Aqib and me. Thank you — questions?"

Stop talking. Smile. Wait.

Q&A routing drill

Rules: Rafay repeats the question → names one person → that person answers in under 30 s → others stay quiet. Never correct a teammate in front of judges.

Topic	First	Backup
System, speed, server, voice, scaling	Abdul Rafay	Moiz
Models, training, YOLO, GPU	Moiz	Abdul Rafay
Data, accuracy, false alarms, privacy	Areeba	Moiz
Website, operator workflow	Aqib	Abdul Rafay
Business, cost, roadmap	Abdul Rafay	Areeba

Drill: one friend reads the questions below in random order. Rafay must route each in under 3 seconds.

#	Judge question	Your 30-second answer (key points)
1	Who did what? (each member answers own part)	Pipeline, async worker, cameras, 12-event engine, risk, audio, voice, evidence, server; connected everyone's work.
2	Why YOLOv8, not RT-DETR?	Fast on a 4 GB GPU; one tool for detection, pose, classification; transformers need more compute.
3	Your fire mAP is only 16% — why?	Hard dataset with tiny fires; every box counts; v2 is 2× v1; alerts decided over time; plan: more epochs, local data.
4	What is precision vs recall?	Precision: when it says fire, how often right. Recall: of all fires, how many found. We prefer fewer false alarms.
5	Hugging or sports = fight?	Body-size distance + fast wrists + repeats 1.5 s + violence classifier ≥ 0.70.
6	Weapon accuracy?	Model works in demo; formal report not finished; I won't give an unproven number.
7	Is this mass surveillance?	Behavioural, not biometric; no face database; human confirms.
8	1000 cameras?	Each camera independent; add edge GPUs; server receives only alerts.
9	Internet goes down?	Detection is local; only Gemini report/voice uses internet, with offline fallback.
10	How is evidence tamper-proof?	SHA-256 fingerprint; one pixel change = different hash.

#	Judge question	Your 30-second answer (key points)
11	How did you get more speed?	Display thread + background specialist thread; 7 → 10 FPS, 48% fewer dropped frames.
12	What is new here?	Many threats + two-stage check + time rules + evidence + routing, for local cities, on a small GPU.

Rehearsal plan, failure drill & score sheet

Day	Practice (مشق)	Goal
Day 1	Read the whole book once. Read the script out loud 3 times.	Know the story
Day 2	Practice with slides and a timer. Record yourself on phone.	Finish on time
Day 3	Practice the live demo 5 times. Break it on purpose once.	Calm demo
Day 4	Q&A drill: a friend asks the 12 questions + 5 random ones.	Answers < 30 s
Day 5	Full run in front of 2–3 people. Fill the score sheet.	Score 4+ everywhere
Final day	One light run only. Sleep well.	Fresh mind

Special team drills

1. **Handover drill** — run only the 4 handover lines, 5 times, until smooth.
2. **Swap drill** — each member presents another member's slides once (for absence / غیر حاضری).
3. **Failure drill** — Aqib stops the backend mid-demo; Rafay keeps talking about slide 7 while Aqib restarts it in under 20 s; if not, play the backup video.

Score sheet (fill for each member)

Skill (1 = weak, 5 = strong)	Run 1	Run 2	Run 3	Run 4
Hook is strong and slow (تیز نہیں)	—	—	—	—
Problem is clear in 40 seconds	—	—	—	—
Demo: says where to look	—	—	—	—
Explains “how it works” simply	—	—	—	—
Says real numbers correctly (no fake numbers)	—	—	—	—
Answers in under 30 seconds	—	—	—	—
Eye contact (آنکھوں سے رابطہ) with judges	—	—	—	—
Finished on time	—	—	—	—
Calm when something breaks	—	—	—	—

Full team guide

Team: **Abdul Rafay** (Lead, AI system) · **Moiz** (ML / model training) · **Areeba** (Data and testing) · **Aqib** (Website and design)

Simple English is used. Hard words have Urdu meaning in brackets, like this: *detect* (بتا لگانا).

Introduction (تعارف)

What is this document?

This is the guide for presenting (پیش کرنا) **VisionGuard ULTRA as a team of four**. It tells you: - who speaks and when, - the exact line to pass the turn to the next person (handover / باری دینا), - who answers which question, - what to do if the demo breaks or someone is absent (غیر حاضر).

Every member must read the full document, not only their own part.

Our project in 30 seconds

VisionGuard ULTRA is an AI system. It makes normal CCTV cameras smart. - It can **detect** (بتا لگانا) fire, fights, guns, big crowds, road accidents, and bags left alone. - It gives a **risk score** (خطرے کا نمبر) from 0 to 100. - It saves the video as **evidence** (ثبوت) that nobody can change secretly. - It sends the alert to the right number: **Police 15, Fire 16, Edhi 115, Rescue 1122**.

Made for the Alibaba AI Hackathon 2026.

How the team introduces itself

Start (Abdul Rafay):

"Assalam-o-Alaikum. We are team **VisionGuard**. We make normal CCTV cameras smart, so they can **find a problem, decide, and send help** very fast. Let us introduce ourselves."

Each person says one line, in this order (5 seconds each, take one small step forward): | Member | Line
| |---|---| | Abdul Rafay | "I am Abdul Rafay, team lead. I built the main AI system." | | Moiz | "I am Moiz, ML engineer. I trained our fire, gun and fight models." | | Areeba | "I am Areeba, data scientist. I prepared the data and I test how correct the models are." | | Aqib | "I am Aqib. I made the website and the control room screen you will see." |

Back to Abdul Rafay: > "Let's start with the problem." → hook.

If time is short (1–3 min): Rafay introduces everyone in one line: "I am Abdul Rafay, with Moiz on models, Areeba on data, and Aqib on the website."

How to use this guide

Section	Who must know it well
1–3 (roles, timing, scripts)	Everyone — your own part word by word, others' parts roughly
4 (questions)	Everyone, mostly Rafay (he gives questions to others)
5 (short time)	Rafay + Areeba (she watches the time)
6 (demo breaks)	Aqib + Rafay
7–8 (practice, small card)	Everyone

0. Team Checklist (one day before + 30 minutes before)

✓ Thing to check	Who
☐ Backend (سروں) running: <code>python prototype/run_web.py</code>	Abdul Rafay
☐ Website running: <code>cd frontend && npm run dev</code> , logged in, tabs in order	Aqib
☐ Model files are there (<code>violence_classifier_best.pt</code> , <code>fire_smoke_best.pt</code> , <code>weapon_detection_best.pt</code>), one test run done	Moiz
☐ Numbers printed from <code>prototype/data_science/ML_Evaluation_Report.md</code> (only real, checked numbers)	Areeba
☐ Backup video (متبادل ویڈیو) of the demo on laptop + USB + phone	Aqib
☐ Full practice at least 3 times with a timer	Everyone
☐ Everyone has read the full document	Everyone
☐ Same type of clothes. Phones on silent.	Everyone

1. Who Does What

Member	Speaks about	Answers questions about	Extra job on stage
Abdul Rafay	Start, problem, solution, system diagram, end	System design, speed, server, voice, many cameras	Moderator (منتظم) — gives each question to the right person
Aqib	Live demo (uses the laptop)	Website, design, how the operator uses it	Demo driver — does all clicks and fixes problems
Moiz	Models and training	Why YOLOv8, training, GPU, speed	Helps with technical questions
Areeba	Data, testing, false alarms, privacy	Data, accuracy, false alarms, bias, privacy	Timekeeper (وقت دیکھنے والی) — shows signal at 1 min and 30 sec left

Where to stand: the speaker stands in the middle front. Aqib stays at the laptop. Others stand a little back and look at the speaker (not at phones).

2. Time Plan (about 7 minutes; for less time see section 5)

Time	Who	Part
0:00–0:30	All (Rafay starts)	Team introduction + hook
0:30–1:00	Abdul Rafay	Problem
1:00–1:30	Abdul Rafay	Solution (Find → Decide → Send help)
1:30–3:30	Aqib	Live demo
3:30–4:30	Moiz	Models and training
4:30–5:30	Areeba	Data, testing, false alarms, privacy
5:30–6:30	Abdul Rafay	System diagram + many cameras + future
6:30–7:00	Abdul Rafay	End → questions

3. What Each Person Says

Abdul Rafay — Start (0:00–1:30)

(First, the team introduction — see Introduction above.)

Hook (ہک — first line to catch attention):

"Karachi has more than 20 million people and thousands of CCTV cameras. But these cameras only **record**. Nobody is really **watching**."

Problem (مسئلہ): - Normal CCTV shows the crime only **after** it happens. - People watching screens get tired after about 20 minutes. One person cannot watch 50 screens. - Help comes late, because first someone must see it, then call. - There are many dangers: fights, guns, fire, big crowds, accidents, bags left alone.

Solution (حل):

"VisionGuard ULTRA makes cameras smart. It watches all cameras together. It finds dangers in **real time** (فوری طور پر), gives a risk score, saves proof, and sends the alert to Police 15, Fire 16 or Rescue 1122. **Find, decide, send help**. Aqib will show you live."

Pass the turn → "Aqib, please show them."

Aqib — Live Demo (1:30–3:30)

#	Click	Say
1	Dashboard (camera grid)	"This is the control room screen. AI is checking every camera live."
2	Play <code>fire.mp4</code>	"It found fire and smoke. It checks for a few seconds first, so there is no false alarm (غلط الارم). Alert goes to Fire Brigade 16."
3	Play <code>fighting.mp4</code> or <code>gun.mp4</code>	"Here is a fight / a gun. You can see the body points (skeleton / ڈھانچہ). Moiz will explain how the system checks a fight."
4	Point at risk score + alert list	"Every event gets a score from 0 to 100. The most dangerous one is always on top."
5	Incidents page	"Every event saves a video clip as evidence. It has a SHA-256 hash — a digital fingerprint (ڈیجیٹل انگوٹھے کا نشان). If someone changes the video, it will not match."
6	(Optional) Say "What is going on?"	"The operator can talk to the system. No clicking needed."

Aqib's rules: tell the judges where to look. Never say sorry for slow speed. Stay calm if something breaks (see section 6).

Pass the turn → "So how does the AI decide this? Moiz."

Moiz — Models and Training (3:30–4:30)

- "The base is **YOLOv8**. It finds objects like people, cars and bags. **YOLOv8-Pose** finds 17 points on the body. **ByteTrack** gives each person a fixed ID number, so we can follow them."
- "Then we **trained** (تربیت دی) **three special models** on Google Colab: one for **fight**s, one for **fire and smoke** (D-Fire data), and one for **guns**."
- "Why YOLOv8? It is fast on a small computer. Our separate-thread design made it 43% faster on a 4 GB Quadro T1000 (7 → 10 FPS at 720p). And one tool can do objects, body points and classification."
- "The video part and the AI part run **separately** (الگ الگ). So the video never freezes (رکتی نہیں)."

Pass the turn → "But a fast model is useless if it gives wrong alarms. Areeba."

Areeba — Data, Testing, Trust (4:30–5:30)

- "First we cleaned the data. We removed broken files, balanced the classes (برابر کیا), and added blur, rain and low light to the images, so they look like real CCTV."
- "False alarms are the biggest risk. So we check **two times**: first fast arm movement starts a check, then the fight model checks the person again. The event also must continue for some time before an alert."
- "We test using precision, recall and mAP (say only the real numbers from the report)."
- Precision* (درستگی) = when it gives an alarm, how often it is right.
- Recall* (پکڑنے کی شرح) = from all real events, how many it catches.
- "About privacy (رازداری): we check **actions**, not **who** the person is. No face recognition, no face database. A human confirms before police are sent."

Pass the turn → "Rafay will show how this works for a whole city."

Abdul Rafay — System, Future, End (5:30–7:00)

Cameras / Mic / SOS → YOLOv8 → ByteTrack → Pose → Special models
→ Event Engine (12 checks over time) → Risk 0–100
→ Send alert · Save evidence · Dashboard · AI short report

- **Many cameras:** each camera works alone. We add more small GPU computers. The main server only gets alerts, not full video.
- **No internet:** detection works on the local computer. Only voice and report need internet, and they have a backup.
- **Future:** test on real city cameras, more local data, use in hospitals, schools and factories, connect directly to helplines.

End (everyone steps forward):

"Cameras are already there. VisionGuard gives them a brain. Find, decide, send help. Thank you. We are happy to take your questions."

4. Team Questions System (سوال و جواب)

4.1 Rules

1. **Abdul Rafay is the moderator.** He repeats the question in simple words (this gives time to think), then says "Moiz, can you answer this?"
2. Only **one person** answers. Others do not interrupt (بیچ میں نہ بولیں). If someone wants to add, wait and say "Can I add one point?" — only one line.
3. Each answer is **less than 30 seconds**.
4. If you don't know, say "Areeba worked on this, she can answer better." — no long silence.
5. **Never say a teammate is wrong in front of judges.** Add gently: "To add to that..."

4.2 Who answers what

Topic	First person	Backup
System design, speed, server, many cameras, voice, Gemini	Abdul Rafay	Moiz
Model choice, training, YOLO vs Transformers, GPU, FPS	Moiz	Abdul Rafay
Data, accuracy, false alarms, bias, privacy	Areeba	Moiz
Website, design, how the operator uses it	Aqib	Abdul Rafay
Business, cost, customers, other products, future	Abdul Rafay	Areeba

4.3 Ready answers (everyone should know all of them)

Question	Simple answer (who)
Why YOLOv8, not RT-DETR / Faster R-CNN?	YOLOv8 is fast and light. Faster R-CNN is too slow for many cameras. Transformers need a strong computer. YOLOv8 does objects + body points + classification in one tool. (Moiz)
What about hugging or sports — false alarm?	Fast movement starts a check → fight model checks again → event must continue for some time → limits can be changed. (Areeba)
How accurate (درست) is it?	Fire model: precision 47.6%, mAP@50 16% — 2× the old colour method; hard dataset with tiny fires; time checks catch events over many frames. Weapon and violence reports still being finished. (Areeba)
How fast?	Measured on Quadro T1000 4 GB: 7 → 10 FPS at 720p (+43%), 48% fewer dropped frames. Target 22+ FPS after tuning. (Moiz)
Can it work with 1000 cameras?	Yes, add more small GPU computers. Main server only gets alerts. (Rafay)
What if internet stops?	Detection works locally. Voice and report have offline backup. (Rafay)
Is this spying (جاسوسی)?	No. We check actions, not faces. No face database. Only logged-in people see data. Evidence has a fingerprint. (Areeba)
What if police are sent by mistake?	The system only suggests. A human checks and confirms. (Rafay)
Is the screen easy to use in stress?	Danger sorted on top, colors, one click to open an event, voice control. (Aqib)
Who will buy it? Cost?	Safe City, police control rooms, malls, universities, hospitals. Uses old cameras, only needs small GPU computers. (Rafay)
What is new in it?	Many dangers + two checks + evidence + alerts in one fast system, made for our cities, works on a small computer. (Rafay)
Who did what?	Each person says their own part in one line (section 1). (All)

4.4 Difficult moments (مشکل حالات)

Situation	What the team does
Nobody knows the answer	Rafay: "Honestly, we have not tested that yet. This is how we would do it..." Never make up numbers.
Judge asks a quiet member directly	That member answers. Others stay quiet, even if they know more. (Judges check if everyone worked.)
Two people start talking	The person who was not given the question stops and says "Go ahead."

Situation	What the team does
Judge doubts (شک) a number	"Fair point. This is from our test computer and videos. Real city test is next."
A teammate says something wrong	Don't correct right away. Later Rafay says: "Just to make that point clear..."
Judge is angry or rude	Rafay answers. Calm, short, says thank you. No arguing (بحث نہیں).
A member is absent	Their backup takes their part (table 4.2). If Aqib is absent, Moiz does the demo. If Rafay is absent, Moiz starts and gives questions.
Time is cut short	Areeba gives the signal → Rafay goes to the End.

5. Short Time? Use This

Time	Who speaks
1 minute	Only Rafay (hook + solution + end). Aqib plays one video quietly.
3 minutes	Rafay (1 min) → Aqib demo (1.5 min) → Rafay end (30 sec). Moiz and Areeba answer questions.
5 minutes	Rafay 1 min → Aqib 1.5 min → Moiz 45 sec → Areeba 45 sec → Rafay 1 min
7+ minutes	Full plan (section 2)

Even in short time: **say all four names**, so judges know who to ask.

6. If the Demo Breaks (Aqib fixes, Rafay keeps talking)

While Aqib fixes it, **Rafay keeps talking** and explains the system diagram. Maximum 20 seconds for each fix.

Problem	Fix	If still broken
Backend stopped	<code>python prototype/run_web.py</code>	Desktop app: <code>python prototype/visionguard_app.py</code>
Website stopped	<code>cd frontend && npm run dev</code> <code>-- --host</code>	Desktop app
Video slow	Use 1×1 layout, close other apps	Simple player: <code>python prototype/run.py --source video --file prototype/Videos/fire.mp4</code>
Mic not working	Type in the voice box	Skip voice
Full crash	—	Play backup video: "This is the same system, recorded before."

7. Practice Plan (مشق کا منصوبہ)

1. **Practice 1 — reading:** everyone reads their part. Use a timer.
2. **Practice 2 — passing turns:** practice every handover line until it is smooth.
3. **Practice 3 — full with fake judges:** a friend asks the questions from 4.3 and some random ones.
Practice giving questions to the right person.
4. **Practice 4 — break the demo:** stop the backend on purpose and fix it.
5. **Swap practice:** each person presents someone else's part one time (in case someone is absent).
6. **English practice:** say your part out loud 5 times. If a word is hard, use a simpler word. Judges care about the idea, not perfect English.

8. Small Card (print one for each person)

```
RAFAY  Introduction → Hook → Problem → Solution → "Aqib, please show them."
AQIB   Dashboard → fire → fight/gun → risk score → Incidents/SHA-256 → voice → "Moiz."
MOIZ   YOLOv8 + Pose + ByteTrack → 3 trained models → 43% faster, separate parts → "Areeba."
AREEBA Clean data → two checks + time check → real numbers → privacy → "Rafay."
RAFAY  System diagram → many cameras / no internet → future → END (all step forward)
Q&A    Rafay gives questions · one person talks · under 30 sec · never make up numbers
BROKEN Rafay talks, Aqib fixes in 20 sec, else backup video
```

Technical knowledge (everyone should know)

— Start to End (پورا پروجیکٹ شروع سے آخر تک)

Abdul Rafay, this part is your **knowledge base** (معلومات کا خزانہ). You will not say all of it in a 3-minute talk. But if a judge asks about **any** part — your part or Moiz's, Areeba's or Aqib's — the answer is here.

A. The whole journey of one video frame (ایک تصویر کا پورا سفر)

Think of the system like a factory line (کارخانے کی لائن). One camera frame goes in, and an alert comes out.

1. CAMERA → frame comes in (webcam / RTSP CCTV / video file)
2. YOLOv8s → finds people, cars, bikes, bags (boxes + confidence)
3. ByteTrack → gives each object a fixed ID and follows it
4. SPECIALISTS → Pose (17 body points), Fire/Smoke, Weapon, Violence, Normal-scene
5. AUDIO → YAMNet listens for gunshot, scream, explosion, glass break
6. EVENT ENGINE → 12 detectors check rules over TIME (not one frame)
7. RISK SCORE → 0–100 → LOW / MEDIUM / HIGH / CRITICAL
8. ACTION → route to department, save evidence (SHA-256), show on dashboard, Gemini writes a short report, operator confirms

Now each step in simple words:

Step 1 — Camera input (کیمرے سے تصویریں) · built by Abdul Rafay - Code: `prototype/camera/` (`webcam.py` , `rtsp.py` , `video_file.py`). - Each camera runs in its own **thread** (الگ کام کی لائن). If an RTSP camera disconnects, it tries to **reconnect** (دوبارہ جڑنا) automatically. - Old frames are dropped, so the system always works on the **newest** frame (no delay building up).

Step 2 — Object detection (چیزیں ڈھونڈنا) with YOLOv8s - Code: `prototype/ai/detector.py` . Model: `yolov8s.pt` (pre-trained on COCO dataset, 80 types of objects). - It draws a **bounding box** (ڈب) around each person, car, bike, bag, and gives a **confidence** (یقین) from 0 to 1. - We keep only boxes with confidence ≥ 0.35 (`DETECTION_CONF_THRESHOLD`). Image size **480** for speed. - **Why YOLO?** "You Only Look Once" — it looks at the whole image one time, so it is very fast.

Step 3 — Tracking (پیچھا کرنا) with ByteTrack - Code: `prototype/ai/tracker.py` . Detection and tracking happen in **one call** (`model.track()`), which doubled the speed. - ByteTrack gives every person a fixed **ID** (e.g. Person #7). It also uses low-confidence boxes to not lose a person in a blur. A lost person is remembered for 30 frames (~1 second). - **Why we need IDs:** to know *how long* someone stands (loitering), *how fast* they move (robbery run), and *who left a bag*.

Step 4 — Specialist models (ماہر ماڈلن) · trained by Moiz, connected by Abdul Rafay - Code: `prototype/ai/specialist_worker.py` . They run in a **separate background thread**, one after another on the GPU. - The worker is **smart** (ہوشیار): it picks what to run based on the scene. Pose runs only when people are present. Fire runs every AI frame (safety first). Weapon runs every 3rd AI frame. - Every result has an **age** (عمر). A result older than 0.75 s is not drawn, older than 1.2 s is not used for events. So old results never cause wrong alerts.

Specialist	What it does	Model
Pose	17 body points (shoulders, elbows, wrists, hips...)	yolov8s-pose.pt (pre-trained)
Fire/Smoke	Boxes around fire and smoke	fire_smoke_best.pt — trained by Moiz
Weapon	Knife, pistol, rifle	weapon_detection_best.pt — trained by Moiz
Violence	Looks at a person crop (کٹی ہوئی تصویر) → violent / not violent	violence_classifier_best.pt — trained by Moiz
Normal scene	Checks if the scene is normal (to stop false alarms)	normal_scene_verifier_best.pt

Step 5 — Audio (آواز) · built by Abdul Rafay - Code: `prototype/ai/audio.py` . Google's **YAMNet** model listens to the microphone in 1-second pieces (16 kHz). - Emergency sounds: gunshot, explosion, scream, crash, glass breaking, siren. Confidence must be ≥ 0.4 . - Sounds from the last 5 seconds are **fused** (ملاپا جاتا ہے) with video. Example: gun seen + gunshot heard = higher risk.

Step 6 — Event Engine (واقعات کا انجن) · built by Abdul Rafay - Code: `prototype/events/engine.py` + one file per event in `prototype/events/` . - **12 detectors**: fire, car accident, bike accident, fight, robbery, kidnapping, weapon threat, crowd crush, loitering, vehicle obstruction, abandoned object, audio emergency. - **How every detector works (same pattern, `events/base.py`)**: 1. Take **signals** (اشارے) from the current frame. 2. Put them in a **time window** (وقت کی کھڑکی) — e.g. last 5–8 seconds. 3. Event must **continue** for at least **1.5 seconds** (`EVENT_MIN_PERSISTENCE_SECONDS`). 4. A short **grace period** (رعایتی وقت) — if YOLO misses one or two frames, the timer does not reset. 5. Calculate **risk score 0–100**. Below 25 = ignored. - To save time, a detector is skipped if it cannot happen (e.g. fight needs at least 2 people).

How the main detectors decide — simple rules:

Event	Rule in simple words
Fight	2+ people are close (distance measured by body size , so it works near and far) + wrists move fast (> 16 px/frame) or arms raised + it repeats for 1.5 s. Violence classifier (≥ 0.70) makes it stronger.
Fire	Fire/smoke confidence ≥ 0.35 , area between 0.2% and 65% of the screen (very big = lighting glitch), and checks if the fire is growing (بڑھ رہی) ($\times 1.3$).
Weapon threat	Weapon box within 250 px of a person, for 0.5 s (very fast because it is very dangerous).
Crowd crush	Number of people vs. zone capacity (50). 70% = warning, 95% = danger.
Car accident	2 vehicles overlap ($\text{IoU} \geq 0.15$) + sudden speed drop (> 15 px/frame).
Bike accident	Rider leans more than 45° + speed drop.
Robbery	Vehicle stops 3+ s, 2+ people run fast (> 30 px/frame) toward someone.

Event	Rule in simple words
Kidnapping	Fast approach + struggle (body points shaking a lot) + forced movement.
Loitering	Person stays in a small area (< 50 px movement) for 5 minutes.
Abandoned object	Bag alone for 60 s, owner more than 200 px away.
Vehicle obstruction	Vehicle stopped (speed < 2) for 10+ s.
Audio emergency	YAMNet emergency sound with confidence ≥ 0.4 .

Step 7 — Risk score (خطرے کا نمبر) | Score | Level | What happens | |---|---|---| | 85–100 | **CRITICAL** | Auto-dispatch suggestion (`RISK_AUTO_DISPATCH_THRESHOLD = 85`) | | 50–84 | **HIGH** | Team review | | 25–49 | **MEDIUM** | Shown as warning | | 0–24 | **LOW** | Ignored |

Step 8 — Action (کارروائی) - Department routing (`config.py` → `DEPARTMENT_ROUTING`):

Event	Department	Dial
Fire	Fire Brigade	16
Fight, robbery, weapon, kidnapping, audio	Sindh Police / Rangers	15
Car / bike accident	Edhi / Rescue	115
Crowd crush	Disaster Management	1122
Vehicle obstruction	Traffic Police	1915
Abandoned object	Police / Bomb Squad	15

- **Evidence (ثبوت)** (`prototype/output/evidence.py`): the system always keeps the last **5 seconds** of video in memory. When an event happens, it saves **5 s before + 3 s after** as a clip, plus a picture. Then it makes a **SHA-256 hash** — a unique digital fingerprint. If even one pixel changes, the hash changes. So nobody can secretly edit the evidence.
- **Database** (`prototype/events/incident_db.py`): every incident is saved in SQLite. It is also used to make **heatmaps (گرم نقشے)** — which areas have most incidents.
- **Gemini report** (`prototype/voice/narrator.py`): turns the event into a short sentence, like "Fight in Sector 4, Police 15 informed."
- **Human confirms**: the operator can **confirm** or **dismiss (رد کرنا)** the alert. AI suggests, human decides.

B. Speed — the async pipeline (الگ الگ چلنے والا نظام — رفتار) • Abdul Rafay

Problem: First version ran all models one after another on the main loop. Video became slow and jumpy.

Solution: We **decoupled** (الگ کیا) it: - **Thread 1 (display):** reads camera, runs fast YOLOv8s + ByteTrack, shows video smoothly. - **Thread 2 (specialist worker):** runs heavy models (pose, fire, weapon, violence) in the background on the newest frame. - AI runs on every 2nd frame (`AI_PROCESS_EVERY_N_FRAMES = 2`). GPU uses **FP16 (half precision / آدھی درستگی)** for speed.

Our real benchmark (`prototype/baseline.json` vs `prototype/async.json` , Quadro T1000, 720p fire video, 300 frames):

	Old (synchronous)	New (async)	Change
Display FPS	7.0	10.0	+43% faster
Dropped frames	866	450	48% fewer
Total time	42.7 s	29.9 s	30% less
GPU memory	144 MB	157 MB	very small

After this, we lowered image size to 480 and added smart scheduling. Our target is 22+ FPS on the T1000 (`config.py`). **Say clearly:** "Our recorded benchmark shows 7 → 10 FPS for the first async version; after more tuning we target 22+ FPS on this small GPU." Do not say 30 FPS unless you show it live.

C. Voice control and backend (آواز سے کنٹرول اور سروس) • Abdul Rafay

- **Voice:** browser listens (Web Speech API, free) → text goes to **Gemini** (`voice/interpreter.py`) → Gemini returns a **JSON command** like `show_cameras` or `status_summary` → `voice/executor.py` does it. If internet is gone, a simple **offline word-matching** (regex) backup understands basic commands.
- **Backend:** `prototype/web/server.py` (FastAPI). Main parts:
 - `/video_feed/{camera_id}` — live video to the browser
 - `/ws/alerts` — **WebSocket:** alerts are **pushed** to the screen instantly (no refresh)
 - `/api/cameras/connect` — add a new camera
 - `/api/evidence/chain/{event_id}` — check evidence hash
 - `/api/heatmaps/karachi` — incident heatmap
 - `/api/login` , `/api/users` — login and users

D. How Moiz made the models (موئز نے ماڈلز کیسے بنائے)

Method — transfer learning (پہلے سے سیکھے ماڈل کو آگے سکھانا): Moiz did not start from zero. He took YOLOv8s, which already knows general shapes from millions of images, and **fine-tuned** (مزید تربیت) it on our special data. This needs less data and less time.

Where: Google Colab, free **T4 GPU**.

Model	Type	Data	Input	Training
Fire/Smoke v2	YOLOv8s detection	D-Fire (~16,000 clean images) + Roboflow fire & smoke	640×640	80 epochs

Model	Type	Data	Input	Training
Weapon	YOLOv8s detection	Weapon CCTV + knife + pistol sets (~7,100 clean images)	640×640	60 epochs
Violence	YOLOv8s-cls classification	RWF-2000 + Real Life Violence + CCTV violence (~12,000 images)	224×224	50 epochs

- **Epoch** (ایک مکمل چکر) = the model sees the whole dataset one time.
- **Detection vs classification:** detection says *where* (box) + *what*. Classification only says *what* for the whole picture (violent / not violent).
- **Why fire v2?** Version 1 used **colour only (HSV)** — red/orange = fire. It gave false alarms on red clothes, sunsets and car lights, and could not see smoke. v2 is a neural network that learned the **shape and texture** of fire and smoke.
- **Two-stage idea for fights:** pose (cheap, fast) first finds suspicious arm movement → only then the violence classifier checks the **person crop**. Max 3 crops per frame to save GPU.

E. How Areeba prepared and tested the data (اریبہ نے ڈیٹا کیسے تیار اور ٹیسٹ کیا)

1. **Cleaning** (صفائی) — `data_science/corrupt_file_scanner.py` - Checks for broken images, empty (0-byte) files, **duplicate** (ایک جیسی) images, and wrong label lines. - **Real results:** D-Fire: 16,027 images, 0 broken, 0 duplicates, 35 bad label lines removed. Weapon: 7,121 images, **5 duplicates removed** → 7,116 clean.
2. **Augmentation** (ڈیٹا بڑھانا) — `data_science/augmentation_pipeline.py` (Albumentations library) - Makes new copies of images with **rain, fog, shadow, brightness change, rotate, zoom**. Real CCTV has bad weather and bad light, so the model must learn these. - **Real result:** 4,304 weather images made.
3. **Class balancing** (برابری) — `data_science/class_balancer.py` - If one class has many more images (e.g. more pistols than knives), the model becomes biased. The script counts classes and **oversamples** (کم والی کلاس کو بڑھانا) the small ones.
4. **False-positive analysis** (غلط الارم کی جانچ) — `data_science/false_positive_analyzer.py` - Looked at 7 saved alerts. Found: crowd crush alarm fired twice in 2 s (duplicate), and 2 fight alarms in **dark frames** (low light). - **Changes applied in config.py** : - Crowd danger 0.90 → **0.95** - Event persistence 1.0 s → **1.5 s** - Detection confidence 0.30 → **0.35** - This is **data-driven tuning** (ڈیٹا کی بنیاد پر بہتری) — we changed limits because of real evidence, not guessing.
5. **Evaluation** (جانچ) — `data_science/model_evaluation.py` - Calculates mAP, precision, recall, confusion matrix and PR curves.

F. Accuracy words — explained simply (درستگی کے الفاظ)

Example: 100 real fires in test images.

Word	Simple meaning	Urdu
TP (True Positive)	Fire was there, model said fire ✓	صحیح پکڑا

Word	Simple meaning	Urdu
FP (False Positive)	No fire, model said fire ❌ (false alarm)	غلط الارم
FN (False Negative)	Fire was there, model missed it ❌	چھوٹ گیا
TN (True Negative)	No fire, model said no fire ✅	صحیح چھوڑا
Precision	When model says "fire", how often is it right? = $TP \div (TP + FP)$	درستگی
Recall	From all real fires, how many did it find? = $TP \div (TP + FN)$	پکڑنے کی شرح
F1-score	One number that balances precision and recall	توازن کا نمبر
IoU	How much the model's box overlaps the real box (0 to 1)	ڈبوں کا ملاپ
mAP@50	Average precision when a box counts as correct if IoU ≥ 0.5	اوسط درستگی
mAP@50-95	Same, but strict — average over IoU 0.5 to 0.95	سخت اوسط درستگی
Accuracy (Top-1)	For classification: % of images with the correct class	درستگی فیصد
Confusion matrix	A table of TP, FP, FN, TN	الجھن کی جدول

Precision vs recall trade-off (سودا): if we raise the confidence limit, we get fewer false alarms (precision ↑) but miss more events (recall ↓). For a city, we prefer **fewer false alarms**, and we fix missed events with **time checks** and **many cameras**.

G. Our real results — say them honestly (ایمانداری سے — اصل نتائج)

Fire/Smoke v2 (from `data_science/ML_Evaluation_Report.md`, D-Fire validation):

Metric	Fire v2 (AI model)	Fire v1 (colour only)	Change
mAP@50	16.0%	~8%	2× better
mAP@50-95	7.6%	~3%	2.5× better
Precision	47.6%	~25%	+90%
Recall	25.5%	~35%	lower (by design: fewer false alarms)

Per class: fire precision **56.1%**, recall 22.9% (2,878 boxes) · smoke precision **39.1%**, recall 28.1% (2,309 boxes).

How to explain these numbers (very important):

"Our fire model doubled the old colour method on mAP and almost doubled precision. The image-level numbers look low because D-Fire is a hard dataset — many fires are tiny, far away or partly hidden, and every single small box is counted. But in our system we don't alert on one frame. The fire must stay for 1.5 seconds and the event engine checks its size and growth. So even if a frame is missed, the event is still caught over many frames. Our next step is more training epochs, a bigger model and local Karachi data to improve these numbers."

Weapon model and violence classifier: training is done / in progress by Moiz, but the **formal evaluation numbers are not finished yet**. Say:

"The weapon and violence models are working in the demo, but their final test report is still being completed. I don't want to give you a number I can't prove."

False alarm rate: target is under 3–5%. It is **still being measured** on a bigger test set. Do not say a fixed % yet.

Sign-off goal (`Accuracy_SignOff_Report.md`): mAP50 $\geq$ 70% for detection, Top-1 $\geq$ 85% for classification. We have **not reached it yet** — this is our honest roadmap.

⚠ Golden rule: **real numbers + honest explanation** wins more trust than big fake numbers.

H. How Aqib made the website (عاقب نے ویب سائٹ کیسے بنائی)

- **Tools:** React + Vite (fast website tools), code in `frontend/src/`.
- **Pages** (`frontend/src/pages/`):

Page	What it shows
<code>Login.jsx</code>	Secure login (only allowed users)
<code>Dashboard.jsx</code>	Camera grid (matrix), live alerts list, risk colors
<code>LiveView.jsx</code>	One camera big, with boxes and skeletons
<code>Incidents.jsx</code>	All past incidents, evidence clips, SHA-256 hash
<code>Registry.jsx</code>	List of all cameras and their locations
<code>ConnectCamera.jsx</code>	Add a new camera (RTSP link / webcam / test video)

- **Components:** `CameraTile.jsx` (one camera box), `AlertList.jsx` (alerts sorted by risk), `LocationTree.jsx` (city → sector → camera).
- **Design ideas:** dark control-room theme (easy for eyes at night), **red / orange / green** colours for risk, most dangerous alert at the top, layouts 1×1 / 2×2 / grid, voice box.

- **How it connects:** video comes from `/video_feed/...`; alerts come live through **WebSocket** `/ws/alerts` — no page refresh needed.

I. Who did what — one-line answers (کس نے کیا کیا)

Member	One line to say
Abdul Rafay (me)	"I designed the full system and built the AI pipeline, async worker, camera input, 12-event engine, risk scoring, audio fusion, voice control, evidence system and FastAPI server, and I connected everyone's work together."
Moiz	"Moiz trained our custom models — fire/smoke, weapon and violence — using transfer learning on Colab T4 GPUs."
Areeba	"Areeba cleaned and checked the datasets, made weather augmentations, balanced classes, analysed false alarms, and ran the evaluation that gave us our metrics."
Aqib	"Aqib designed and built the React command-center website — dashboard, live view, incidents, camera registry and voice box."

J. Weak points — say them before the judge does (کمزوریاں)

Judges respect a team that knows its limits (حدود):

1. Fire model numbers are still low on the hard D-Fire test → plan: more epochs, bigger model, local data.
2. Weapon and violence final reports are not finished → plan: run `model_evaluation.py` for both.
3. Small weapons far from the camera are hard to see → plan: higher resolution crops near people.
4. Dark / night video gives more false fights → plan: low-light augmentation and IR cameras.
5. Not yet tested on a real city camera network → plan: pilot project.
6. Pixel rules (like 200 px distance) depend on camera position → plan: per-camera settings (calibration / ترتیب).